Classwork2

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You are interested in the relation between sold quantities and prices and for that, you collect data on sales from a famous online store.

1. Create a Word file named `yourname_yoursurname_cw2.docx`
2. Create a new do-file and save it as `yourname_yoursurname_cw2.do`
3. Set the working directory
4. Import the database named "summer_products_with_rating.csv" in Stata.
5. Plot the distribution of the variables `units_sold` and `price`. Include the graphs in the Word file.
6. Estimate the following regression:
$$units\_sold_i = \beta_0 + \beta_1 price_i + \varepsilon_i \tag{1}$$
Copy-paste the output in the Word file.
7. Based on regression (1) should you conclude that the price affects the sold quantities? Why?
8. Now, you suppose that the relation between the two variables may not be linear. So you decide to include in the regression the square of price:
$$units\_sold_i = \beta_0 + \beta_1 price_i + \beta_2 price_i^2 + \varepsilon_i \tag{2}$$
Copy-paste the output in the Word file.
9. The two variables are jointly statistically significant? Include the output in the Word file and briefly comment on it.
10. Now, you realize that it would be more appropriate to estimate the elasticity between the two variables. Compute the appropriate regression. Copy-paste the output in the Word file.
11. You want to control in your regression for the *quality of the product*. For this reason, you decide to add to the regression estimated at the previous point the variable `rating`. Copy-paste the output in the Word file. Do the results suggest that the omission of the variable `rating` generates *omitted variable bias*?

12. To the regression estimated in the previous point add, in the appropriate way, the control for the size of the item (variable `size`). Copy-paste the output in the Word file.

13. Using the command `testparm` jointly test the significance of the coefficients related to *size*. Copy-paste the output in the Word file.

14. Based on the results obtained in the previous points, would you conclude that the demand for the analyzed goods is elastic or inelastic with respect to price?